

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE)

May 2012

CE202 DATA COMMUNICATION & NETWORKING

Date: 02.05.2012, Wednesday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

- Q1 Answer the following questions. [11]**
- [A] Define: bit rate, bit interval, digital signal, analog signal [2]
 - [B] What are the advantages of distributed processing? [2]
 - [C] What is Critical Angle? [2]
 - [D] Draw the RZ encoding signal for 01001110 bit pattern. [2]
 - [E] What is simplex, Half-duplex and Full-duplex communication? [3]
- Q2 Answer the following questions. (Attempt any Three) [12]**
- [A] What are the advantages and disadvantages of optical fiber?
 - [B] How light signals are passed through fiber cable? What are the propagation modes?
 - [C] Explain Manchester and differential Manchester encoding method by taking suitable example.
 - [D] Write and explain any four types of propagation of radio waves.
 - [E] Explain NRZ-L and NRZ-I encoding method.
- Q3 Answer the following questions.(Attempt any Two) [12]**
- [A] Draw various topologies and explain advantages & disadvantages of each.
 - [B] Write down the layers of OSI model and explain functionality of each layer.
 - [C] Write a short note on PCM(Pulse Code Modulation).

SECTION-II

- Q4 Answer the following questions. [11]**
- [A] Define framing and Interleaving with respect to TDM. [2]
 - [B] Write full form of: DSL, ADSL, HDSL, VDSL [2]
 - [C] Write down fields of Ethernet frame format. [2]
 - [D] Write down differences between LRC & VRC. [2]
 - [E] Differentiate LAN, MAN & WAN. [3]
- Q5 Answer the following questions. (Attempt any Three) [12]**
- [A] Explain in detail : Distance Vector Routing Algorithm
 - [B] Write a short note on Analog Switched Service.
 - [C] How Selective Repeat is better than Go-back-N error control method?
 - [D] Explain Poll & Select method of Line discipline.
 - [E] What is full form of CSMA/CD? Write down mechanism of CSMA/CD.
- Q6 Answer the following questions.(Attempt any Two) [12]**
- [A] Write a short note on: FDM, TDM and WDM.
 - [B] By taking suitable example, explain how CRC is computed.
 - [C] What are the functionalities of Repeater, Bridge, Switch and Router?